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Federated transfer learning for generalized salt body segmentation across multiple heterogeneous seismic datasets

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Abstract

Seismic segmentation is a key step in subsurface interpretation and hydrocarbon exploration, where precise delineation of features such as salt bodies guides reservoir modeling and drilling decisions. Progress in deep learning, particularly encoder-decoder architectures like U-Net, has improved segmentation accuracy. Yet, model development remains constrained by the scarcity of labeled seismic data and strict data privacy requirements. In this study, we propose a novel TransFed-SaltNet model that addresses these challenges by presenting a privacy-preserving framework that combines federated learning with transfer learning. The approach utilizes pre-trained EfficientNet-B7 encoders with U-Net decoders and fine-tunes them on distributed seismic datasets, enabling collaborative training without sharing raw data. Experiments on salt body segmentation tasks show that the TransFed-SaltNet achieves segmentation performance within 2–3% of a centrally trained model. It also demonstrated a 9.3% increase in the IoU compared to scratch-trained models. These results highlight the model's exceptional robustness and generalizability on unseen held-out data despite training on highly heterogeneous seismic datasets, offering a promising balance between performance and data confidentiality.

Keywords Salt, Seismic, Segmentation, Deep learning, Transfer learning, Federated learning

1 Introduction

In seismic interpretation, salt domes play a critical role because they act as traps for oil and gas resources, making them highly significant for the hydrocarbon industry [1]. Traditionally, delineation of salt boundaries relied on manual interpretation, which was time-consuming and subject to interpreter bias. Later, machine learning methods were introduced to automate the process, offering partial relief but often struggling with complex geological features [2]. Recent advances in deep learning, particularly the use of U-Net-based architectures [3] and transfer learning, have substantially improved segmentation performance and enabled more robust and accurate detection of salt structures.

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Despite their success, deep learning models are notoriously data-hungry, requiring vast amounts of high-quality, labeled data for training—a– requirement often unattainable in seismic interpretation due to the scarcity of expert-annotated data and the immense cost of acquisition. Furthermore, the oil and gas industry operates under stringent data privacy and confidentiality regulations, making it prohibitively difficult for organizations to pool their seismic data into a central repository for model training [4]. This creates a fundamental dilemma: how to develop a robust, generalizable model without direct access to distributed and confidential datasets.

Federated Learning (FL) presents a paradigm shift to address this privacy bottleneck. It is a decentralized machine learning approach where a global model is collaboratively trained across multiple clients (e.g., different energy companies) without any raw data ever leaving its original source [5]. Instead of sharing data, clients train a model locally on their private datasets and only share the model's trained parameters (e.g., gradients or weights) with a central server for aggregation. This framework ensures data privacy and security while still leveraging the collective knowledge embedded in heterogeneous datasets from various sources. However, a key challenge in a federated setting is that training often starts from a randomly initialized model, which can lead to slow convergence and suboptimal performance, especially when the client datasets are small.

To simultaneously tackle the challenge of limited data, Transfer Learning (TL) offers a solution. TL involves leveraging a model pre-trained on a large, general-purpose dataset (e.g., ImageNet) and fine-tuning it for a specific downstream task, such as seismic segmentation. This approach is highly beneficial as it allows the model to inherit rich, generic feature detectors from the source domain, significantly reducing the amount of task-specific labeled data required for effective training and improving convergence speed. A pre-trained encoder, such as EfficientNet-B7, can provide a strong foundational understanding of visual features, which can then be adapted through fine-tuning to the specific textural and structural patterns of seismic images [6].

In this paper, we propose a novel framework, TransFed-SaltNet, that combines Federated Learning with Transfer Learning to create a robust and privacy-preserving solution for salt body segmentation. Our method employs a U-Net architecture with a pre-trained EfficientNet-B7 encoder, which is collaboratively fine-tuned across multiple institutional clients holding heterogeneous seismic datasets. This approach ensures data confidentiality while harnessing the collective power of distributed data to build a generalizable model. Experimental results demonstrate that the TransFed-SaltNet model achieves performance comparable to a model trained on centralized data, highlighting its practical potential for real-world seismic interpretation challenges.

1.1 Contributions

Some of the main contributions of this study are given below.

- Introduced a privacy-preserving federated segmentation framework, TransFed-SaltNet, that enables collaborative seismic model training across organizations without data sharing, unlocking new possibilities for real-world deployment under strict confidentiality.
- Integrated a pre-trained EfficientNet-B7 encoder within U-Net to boost feature extraction and achieved near-centralized segmentation accuracy within only 3–5% IoU despite training on highly heterogeneous seismic datasets.

- Demonstrated exceptional generalizability and robustness across independent test sets, proving the practicality of federated learning for scalable, high-performance seismic interpretation under real-world constraints.

2 Literature review

Recent research on seismic salt body segmentation shows a clear shift from manual interpretation to deep learning approaches, with U-Nets leading performance gains. This section first reviews these methods, then highlights the persistent problem of limited labeled seismic data and emerging solutions. It discusses data augmentation as a key strategy to enhance model training, followed by advances using vision transformers and foundation models. Finally, it covers privacy-aware approaches such as federated learning and recent efforts combining federated and transfer learning for robust, distributed seismic interpretation.

2.1 Deep learning for salt body segmentation

Traditional methods for salt body delineation relied heavily on manual interpretation or basic image processing techniques, which were often time-consuming and error-prone. The advent of deep learning, particularly Convolutional Neural Networks (CNNs), brought a significant breakthrough in automating salt segmentation tasks. Architectures like U-Net and its variants have demonstrated impressive performance in learning spatial hierarchies from seismic data, outperforming classical methods in accuracy and efficiency [3]. Nevertheless, these models often face generalization challenges when deployed on datasets from different basins or surveys, primarily due to domain shifts and overfitting to limited training data [7].

2.2 Data scarcity in seismic segmentation and emerging solutions

A major bottleneck in applying deep learning to seismic interpretation is the scarcity of labeled data. Seismic data labeling requires domain expertise, and manual annotation is not only labor-intensive but also lacks scalability [2]. To mitigate this, various strategies such as self-supervised learning, semi-supervised learning, and transfer learning have been explored. Self-supervised techniques, including contrastive learning and masked autoencoding, enable models to learn useful representations from unlabeled data [8]. Semi-supervised methods leverage a small labeled set alongside a larger unlabeled pool, achieving better generalization with minimal annotation effort [9]. Transfer learning has shown promise in seismic tasks by adapting pretrained models (often from natural images or domain-specific pretraining) to new geological settings, thereby reducing the demand for large labeled datasets [10]. Multi-Task Learning (MTL) [11] has emerged as significant paradigm for generalized seismic interpretation. MTL improves generalization by learning shared representations across multiple heterogeneous datasets simultaneously. However, MTL typically requires centralized data pooling, which is often prohibited by the strict data privacy policies of energy companies. Few-shot learning (FSL) methods [12] address limited labeling but often struggle to capture the full statistical complexity of large, decentralized datasets. While FSL provides quick adaptation, it fails to leverage the extensive private data volumes siloed within different organizations.

2.3 Data augmentation techniques

To further address data limitations, data augmentation is widely adopted in seismic learning pipelines. Conventional augmentation methods include flipping, rotation, scaling, and adding noise to simulate variability and prevent overfitting [2]. More advanced techniques utilize generative models, particularly Generative Adversarial Networks (GANs), to synthetically generate realistic-looking seismic images and masks. GAN-based augmentation not only expands dataset diversity but also introduces novel geological variations that may not exist in the original data [13].

2.4 Recent advances: vision transformers and foundation models

Recent advancements in deep learning for seismic interpretation have seen a growing interest in Vision Transformers (ViTs) and foundation models as promising alternatives to traditional CNN-based architectures. Transformers, originally developed for natural language processing, have been adapted for vision tasks due to their ability to model long-range dependencies and global context. In the geophysical domain, [14] proposed SaltFormer, a hybrid CNN-Transformer architecture tailored for automatic detection of salt dome. By combining the local feature extraction strength of CNNs with the global attention mechanism of Transformers, SaltFormer achieved improved accuracy and robustness in challenging seismic settings. This highlights the potential of Transformer-based models to better capture the irregular shapes and variable textures of salt bodies, which are often difficult for traditional CNNs to model effectively.

In parallel, the concept of foundation models has emerged as a new paradigm in machine learning. These models are typically pre-trained on large-scale datasets and designed to generalize across a wide range of downstream tasks with minimal fine-tuning. Gao et al. [15] introduced a foundation model driven by a multi-modal prompt engine, capable of performing universal geobody interpretation across different surveys. Although multimodal in design, the core strength of the model lies in its foundation-level adaptability, allowing it to generalize across data distributions and geological contexts.

2.5 Federated learning in geoscience

Federated Learning (FL) is an emerging paradigm designed to facilitate collaborative model training across decentralized data sources without transferring raw data [5]. In geoscience, where datasets are often distributed across companies, institutions, or regions with privacy and licensing restrictions, Federated Learning presents an attractive solution [16]. Although FL has seen growing adoption in healthcare and finance, its application in geophysics is still nascent. FL enables learning from diverse datasets while preserving data locality, but challenges such as non-IID data distributions, communication efficiency, and convergence stability are non-trivial in practice. Nonetheless, the potential of FL in enabling secure, collaborative learning for seismic interpretation is substantial. ul Islam and Wali [17] proposed GeoFedNet, a privacy-aware framework for various seismic interpretation tasks, and demonstrated the applicability of federated learning on seismic interpretation. Although their results are close to centralized training, the models used were trained from scratch and lack the robustness that can be leveraged through the use of transfer learning.

2.6 Combining federated and transfer learning

Wali et al. [6] applied transfer learning for salt segmentation and reported improved performance compared to training from scratch. Integrating FL with transfer learning represents a powerful synergy for seismic applications. While FL enables data collaboration without centralization, transfer learning improves model adaptation across heterogeneous datasets and geological environments. Recent studies have demonstrated that combining the two can enhance model robustness, especially in scenarios with distribution shifts and limited annotations [18]. In the context of salt body segmentation, such a hybrid approach could allow institutions to collaboratively train high-performing models while respecting data privacy, ultimately leading to more generalized and reliable salt detection systems across different basins.

3 Methodology

The methodology section details the structured approach adopted for salt dome segmentation in seismic images using federated learning. It begins with Data Retrieval and Preprocessing, which includes a description of the datasets used and the preprocessing steps applied to ensure uniformity and quality. The Model Architecture subsection outlines the encoder-decoder design chosen for segmentation. The Federated Learning Process describes how model training was distributed across clients without sharing raw data. This is followed by the Experimental Setup, which includes configuration parameters and training details. Finally, Model Evaluation explains the metrics used to assess segmentation performance. The overall workflow is illustrated in Fig. 1.

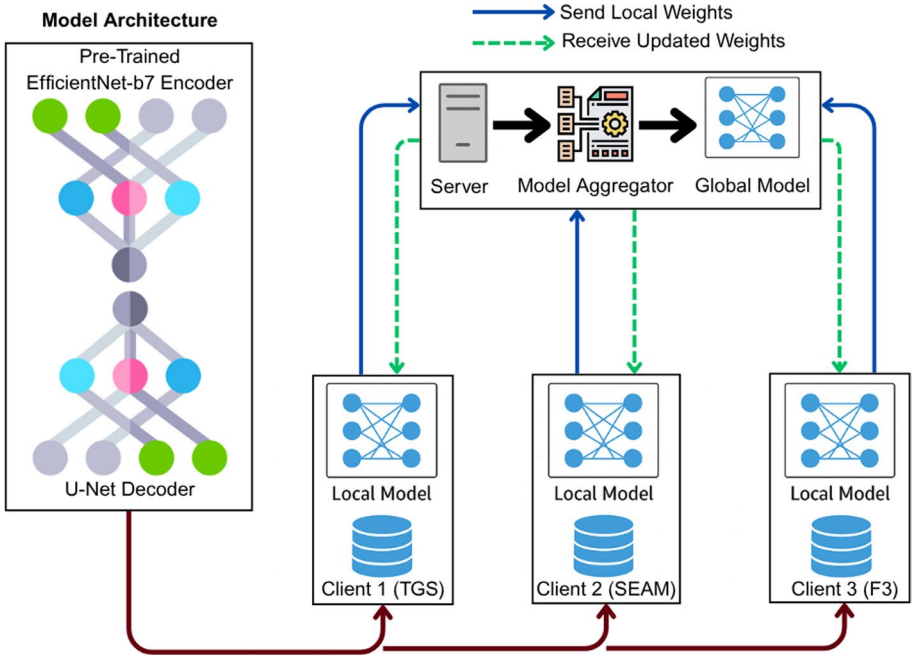


Fig. 1 TransFed-SaltNet Architecture: Federated learning setup with three clients——each using a unique seismic dataset (TGS, SEAM, and F3)——and a central server. Each client employs a shared encoder-decoder architecture: an EfficientNet-B7 encoder pretrained on ImageNet and a U-Net-style decoder. The blue solid lines represent the transmission of local model weights to the server, while the green dotted lines indicate the aggregated weights sent back to the clients by the server

3.1 Data retrieval and preprocessing

In this section, we describe the datasets used in our experiments and detail the preprocessing steps undertaken to prepare the data for training and evaluation within a federated transfer learning framework.

3.1.1 Dataset

In this study, we simulate a federated learning environment by assigning one dataset to each client, resulting in a three-client setup. Each client holds data from a distinct publicly available seismic dataset: TGS, SEAM, and F3. This setup naturally introduces heterogeneity in terms of data distribution, image resolution, labeling granularity, and acquisition domains, making it a suitable testbed for federated segmentation.

Derived from the TGS Salt Identification Challenge [19], this dataset consists of 4000 grayscale seismic images of size 101×101, each with a binary salt mask. Additionally, depth metadata is provided for each image. The official test set includes unlabeled images and is not used for training. Due to its smaller patch size and high foreground-background variability, this dataset introduces significant class imbalance at the client level.

The SEAM dataset [20] contains pixel-level annotations for salt and non-salt regions across three seismic inlines (4403, 4499, and 4595). These inlines were divided into smaller overlapping patches to serve as training samples. Compared to TGS, SEAM features higher resolution and more continuous salt structures, providing a distinct spatial and labeling distribution.

The F3 dataset [21] includes four labeled inlines (190, 290, 390, and 490) extracted for salt segmentation. Originally annotated for 13 different seismic structures, we binarized the labels to indicate salt vs. non-salt regions for consistency. This dataset offers large, complex patterns and significant structural diversity, further amplifying the non-IID nature of the federated setup.

Table 1 provides an overview of the training data used by each client in the federated setup. The datasets vary in size and class distribution. Client 1 uses the TGS dataset with 3,400 samples, containing approximately 25% salt (foreground) pixels. Client 2 utilizes the SEAM dataset with 3,196 samples and a comparatively higher foreground ratio of 29%. Client 3 is assigned the F3 dataset, comprising 4,350 samples, but with a significantly lower foreground proportion of only 1.5%. These variations reflect the natural class imbalance commonly encountered in seismic datasets.

3.1.2 Data preprocessing

The preprocessing pipeline involved a series of steps to standardize the data and prepare it for training under a federated transfer learning framework. Each dataset was processed independently at the client level, following a consistent methodology.

Table 1 Client-wise training dataset distribution showing the number of samples and foreground (salt) vs. background (no-salt) ratios

Client	Dataset	Training samples	Foreground (%)	Background (%)
Client 1	TGS	3,400	25.0	75.0
Client 2	SEAM	3,196	29.0	71.0
Client 3	F3	4,350	1.5	98.5

First, all images and their corresponding binary salt masks were loaded into a Pandas DataFrame. Pixel values were normalized to the [0,1] range by dividing by 255. Additionally, salt coverage for each sample was computed as the ratio of salt pixels to the total number of pixels in the mask. This metric was later used for stratified splitting to ensure balanced representation of salt presence across data splits.

For the TGS dataset, 20% of the samples were randomly reserved for testing. In contrast, for the SEAM and F3 datasets, specific inline sections were held out to serve as the test set——inline 4403 for SEAM and inline 190 for F3——while the remaining inlines were used for training. From the training portion of each dataset, a stratified 20% subset was extracted to serve as the validation set, using salt coverage as the stratification criterion to preserve class distribution.

All images and masks were resized to a uniform shape of $128\times128\times3$ to meet the input requirements of the encoder backbone used in transfer learning. This resizing also ensured consistency across clients, despite the original datasets having different resolutions and formats.

3.2 Model architecture

In this study, we adopt an encoder-decoder architecture tailored for binary image segmentation. The encoder is based on EfficientNet-B7, a convolutional neural network pretrained on ImageNet, selected for its strong capability in extracting high-level semantic features while maintaining computational efficiency [22]. Leveraging a pretrained encoder accelerates convergence and enhances performance, particularly when training on limited domain-specific data such as seismic images.

The decoder follows the U-Net architecture, which is characterized by a symmetric, U-shaped design with skip connections linking corresponding layers in the encoder and decoder paths. These skip connections play a crucial role in segmentation tasks by combining low-level spatial information from early encoding layers with the high-level semantic features obtained during decoding. This fusion allows the network to recover fine-grained details and achieve accurate boundary delineation, which is essential for precise pixel-wise classification.

Together, the EfficientNet-B7 encoder and U-Net-style decoder form a robust architecture capable of capturing both global context and fine spatial details, making it well-suited for the challenges of seismic image segmentation.

3.3 Federated learning process

In our federated learning setup, as illustrated in Fig. 1, we utilize a client–-server architecture involving three independent clients, each associated with a unique seismic dataset: TGS, SEAM, and F3. The key goal is to collaboratively train a shared deep learning model without transferring any raw data between clients and the central server, thereby preserving data privacy and security.

Algorithm 1 Federated averaging with federated validation (FedAvg-FV)

Input: Number of clients K, communication rounds T, local epochs E, learning rate η
Initialize: Global model weights w_0, best IoU $IoU_{best} \leftarrow 0$, best weights $w^* \leftarrow w_0$
for each round $t = 0, 1, \ldots, T-1$ **do**
  for each client $k = 1, 2, \ldots, K$ **in parallel do**
    $w_{k,t} \leftarrow w_t$ {Download current global model}
    Local Training:
    for each local epoch $e = 1, \ldots, E$ **do**
      Update $w_{k,t} \leftarrow w_{k,t} - \eta \nabla \ell(w_{k,t}; \mathcal{B}_k)$ using local data $\mathcal{B}_k$
    end for
    Send updated local weights $w_{k,t}$ to server
  end for
  Server Aggregation:
  $w_{t+1} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} w_{k,t}$ where $n = \sum n_k$
  Federated Validation:
  for each client $k = 1, 2, \ldots, K$ **in parallel do**
    $IoU_{k,t+1} \leftarrow \text{LocalEvaluate}(w_{t+1}, \mathcal{V}_k)$ {Evaluate on local validation set}
  end for
  $IoU_{t+1} \leftarrow \sum_{k=1}^{K} \frac{v_k}{V} IoU_{k,t+1}$ {Weighted average of metrics where $V = \sum v_k$}
  if $IoU_{t+1} > IoU_{best}$ **then**
    $IoU_{best} \leftarrow IoU_{t+1}$
    $w^* \leftarrow w_{t+1}$ {Update best model checkpoint}
  end if
end for
Output: Best global model weights w^*

Each client is equipped with an identical encoder-decoder network architecture composed of an EfficientNet-B7 encoder pretrained on ImageNet and a U-Net-style decoder. This shared architecture ensures compatibility of model parameters across clients and enables effective aggregation at the server. During training, each client performs computation locally using its own private seismic dataset, which introduces natural heterogeneity in the data distribution across clients——a typical characteristic in federated learning scenarios.

The federated training process follows the Federated Averaging with Federated Validation (FedAvg-FV) algorithm (1), where training is conducted over multiple communication rounds. At the beginning of each round *t*, the central server broadcasts the current global model parameters w_t to all clients. Each client then initializes its local model with the received global weights and proceeds to train the model on its local dataset for a fixed number of epochs *E* using stochastic gradient descent. Upon completing the local training phase, the clients transmit their updated local model weights $w_{k,t}$ back to the central server. These model updates are represented by the solid blue lines in Fig. 1.

Once all client updates are received, the server performs an aggregation step, computing a weighted average of the client models based on the number of training samples n_k at each client to produce the new global state w_{t+1}. To maintain strict data privacy, a federated validation step is performed immediately following aggregation. Rather than evaluating the model on a centralized dataset, the server redistributes w_{t+1} to the clients, who evaluate the model on their locally reserved validation sets $\mathcal{V}_k$. Clients transmit only the resulting performance metrics back to the server. The server then computes a global mean Intersection over Union (mIoU) by calculating a weighted average of these local metrics. If the current round produces a higher global mIoU than previously observed, the model weights are saved as the best global model checkpoint w^*.

The updated global model is subsequently redistributed to all clients for the next round of training, as indicated by the green dotted lines in Fig. 1. This cycle of local training, weight uploading, server-side aggregation, and global model redistribution continues for a predefined number of communication rounds T. At the conclusion of the process, the algorithm outputs the best-performing weights w^* rather than the final round's weights, ensuring optimal segmentation performance.

Besides FedAvg, this study implements FedProx [23] as a comparative baseline to evaluate performance under heterogeneous data conditions. FedProx introduces a proximal term to the local objective function to restrict local updates from deviating significantly from the global model. The local optimization objective for client *k* is defined as:

$$min_w\, h_k(w, w^t) = F_k(w) + \frac{\mu}{2}\|w - w^t\|^2$$

where $F_k(w)$ represents the local loss and μ controls the regularization strength. This comparison identifies whether rigid regularization improves global stability or hinders local adaptation in seismic salt segmentation. For our experiment, μ was set to 0.01 for the first 100 iterations and then lowered to 0.001 for the next iterations.

By adopting this federated learning framework, the model benefits from the diversity of all three seismic datasets while maintaining the privacy of each client"s data across both training and validation phases. The use of a strong, pretrained encoder (EfficientNet-B7) combined with this collaborative learning and checkpointing paradigm enables the model to generalize better across varying geological contexts present in the different datasets.

3.4 Experimental setup

To simulate federated learning, we developed a custom implementation from scratch using TensorFlow. The simulation involves three clients, each associated with a distinct seismic dataset. The model architecture deployed across all clients follows an encoder-decoder structure, consisting of an EfficientNet-B7 encoder pretrained on ImageNet and a U-Net-style decoder. The encoder and decoder were integrated using components from the segmentation-models library to ensure consistency and scalability in segmentation tasks. Encoder-Decoder combination was fined tuned during training process to effectively capture survey specific geological signatures.

Each client trains its local model independently using its private data for one epoch per communication round. After local training, the updated model weights from all clients are transmitted to a central server, which performs federated averaging. Specifically, a weighted average is computed based on the number of training samples per client to update the global model. This global model is then redistributed to all clients for the next round of training. The process repeats for a total of 250 communication rounds.

Training utilized the Stochastic Gradient Descent (SGD) optimizer with a fixed learning rate of 0.001 and a momentum of 0.9. We did not apply weight decay because the limited local training (one epoch) inherently mitigates local overfitting. The batch size was 32. To address class imbalance, we adopted a composite loss function combining Binary Cross-Entropy and Dice loss with equal weighting (1:1). All experiments used a fixed random seed of 42 to ensure reproducibility. Validation was conducted using a federated validation protocol with locally held-out splits. Model performance was

monitored using the Intersection over Union metric. The best-performing global model based on the aggregated validation IoU was saved at each round.

All models were trained and evaluated using GPU acceleration to reduce training time. The federated learning framework was implemented without the use of external federated libraries, providing full control over the training pipeline and aggregation logic.

3.5 Model evaluation

The trained models were evaluated on the validation and test sets using key metrics, including Intersection over Union (IoU), precision, recall, F1-score, and accuracy. IoU served as the primary evaluation metric, with a thresholding step applied to convert the model''s continuous probability outputs into binary masks. To optimize IoU, thresholds ranging from 0.3 to 0.7 were tested in increments of 0.01, selecting the threshold that maximized IoU on the validation set.

Beyond IoU, additional metrics such as precision, recall, and F1-score were calculated to provide a balanced assessment of segmentation performance, highlighting the trade-off between over-segmentation (low precision) and under-segmentation (low recall). Accuracy was also computed to evaluate overall classification effectiveness. This comprehensive approach ensured a robust evaluation of the model''s performance across diverse metrics, complementing the primary IoU analysis.

4 Results and discussion

We compared the TransFed-SaltNet framework against a centralized Multi-Task Learning (MTL) baseline. This model uses pooled data from all clients to establish an upper performance bound for the federated system. Both models share the EfficientNet-B7 and U-Net architecture. The Centralized Model uses standard supervised training on pooled data without privacy. TransFed-SaltNet maintains data locality and uses FedAvg for privacy-preserving aggregation. The evaluation metrics include IoU, Precision, Recall, F1-score, and Accuracy. The results are summarized in Table 2.

4.1 Global performance: centralized vs. federated benchmarking

The centralized model was trained with direct access to the pooled training data from all three clients. This setup provides the highest overall performance because it benefits from full data diversity and volume without the constraints of privacy-preserving protocols. On the held-out test set, the centralized model achieved an IoU of 0.922, a precision of 0.9716, a recall of 0.9473, an F1-score of 0.9593, and an accuracy of 0.9866. These results serve as the performance upper bound for the study. They provide a critical baseline for evaluating the privacy-utility tradeoff inherent in the federated framework.

The federated framework (TransFed-SaltNet) enables collaborative training while keeping raw seismic data local. The model achieves competitive performance using a federated validation protocol. These metrics represent the weighted average of results

Table 2 Performance comparison of centralized and TransFed-SaltNet models using common evaluation metrics

Model	Dataset	IoU	Precision	Recall	F1-score	Accuracy
Centralized Model	Testing (Held-out)	0.922	0.9716	0.9473	0.9593	0.9866
TransFed-SaltNet	Federated Validation	0.899	0.9661	0.9055	0.941	0.975
TransFed-SaltNet	Testing (Held-out)	0.874	0.9826	0.8504	0.9117	0.9726

calculated independently on local client splits. On the federated validation set, the model produced an IoU of 0.899, precision of 0.9661, recall of 0.905, F1-score of 0.941, and accuracy of 0.975. The high precision and segmentation quality confirm effective knowledge transfer across diverse geological domains despite the data remaining isolated.

Evaluation on held-out test data assesses the generalization capability of the federated model. The model achieved an IoU of 0.874, precision of 0.9826, recall of 0.8504, F1-score of 0.9118, and accuracy of 0.9726. These results are further illustrated in Fig. 2, which provides a clear visual comparison of both models using two key segmentation metrics. These results confirm the robustness of the framework when processing unseen seismic features. Performance shows only a minor decrease compared to validation results and remains competitive with the centralized training baseline. These findings demonstrate that the model generalizes effectively across the heterogeneous geological distributions represented in the surveys.

4.2 Per-client performance comparison

Individual client performance is evaluated across four distinct training configurations: Local-Only, FedAvg, FedProx and centralized MTL. This granular analysis demonstrates how federated collaboration impacts local utility compared to independent training. By assessing each survey separately, the study identifies whether global knowledge transfer successfully mitigates local challenges such as extreme class imbalance and domain heterogeneity. These metrics reveal the specific performance trade-offs between different federated optimization strategies. The performance metrics in Table 3 confirm that federated frameworks effectively manage heterogeneous seismic data. FedAvg provides a universal performance uplift across all clients. Client 3 (F3) experiences the most significant recovery. Its F1-score rises from 0.7510 to 0.9118. This improvement proves that the collaborative process successfully transfers robust salt features from data-rich domains to address extreme class imbalance. Client 2 (SEAM) also achieves its highest IoU of 0.945 under this protocol. These findings indicate that FedAvg allows for necessary local adaptation while benefiting from a shared knowledge base.

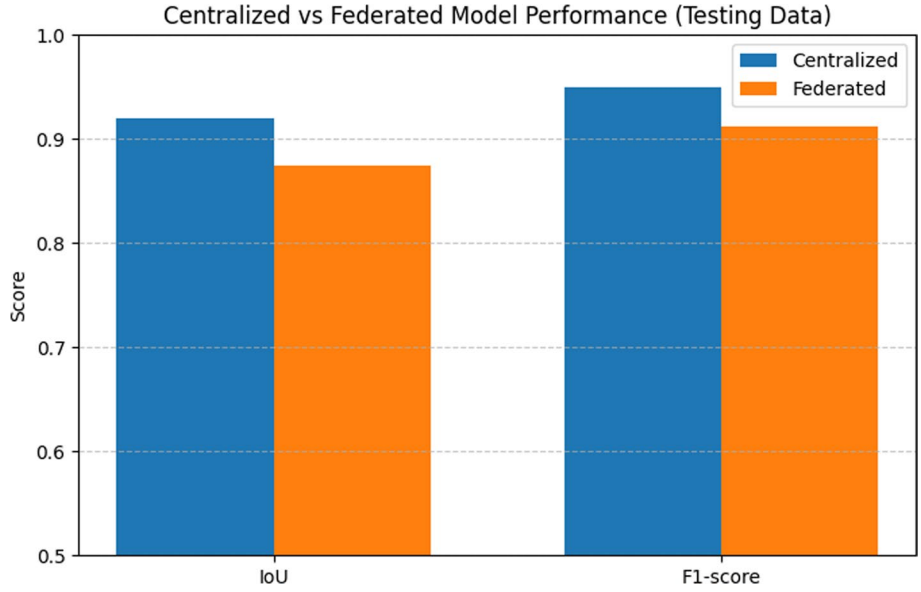


Fig. 2 Comparison of Centralized and Federated model performance on testing data in terms of IoU and F1-score

Table 3 Per-client validation performance comparison

Client (Dataset)	Training setup	IoU	F1-score	Accuracy
Client 1 (TGS)	Local-Only	0.853	0.9320	0.9720
	FedAvg	0.877 (↑)	0.9514 (↑)	0.9841 (↑)
	FedProx	0.851 (↓)	0.9340 (↑)	0.9810 (↑)
	Centralized (MTL)	0.8791	0.9530	0.9851
Client 2 (SEAM)	Local-Only	0.932	0.9705	0.9810
	FedAvg	0.945 (↑)	0.9802 (↑)	0.9895 (↑)
	FedProx	0.933 (↑)	0.9750 (↑)	0.9850 (↑)
	Centralized (MTL)	0.9460	0.9831	0.9900
Client 3 (F3)	Local-Only	0.901	0.7510	0.9710
	FedAvg	0.924 (↑)	0.9118 (↑)	0.9823 (↑)
	FedProx	0.891 (↓)	0.8110 (↑)	0.9700 (↓)
	Centralized (MTL)	0.9310	0.9211	0.9825

Arrows indicate improvement (↑) or decline (↓) relative to the Local-Only baseline

FedProx demonstrates a different performance profile. While it improves F1-scores compared to local-only training, it results in a decrease in IoU for Client 1 and Client 3. Specifically, Client 3 sees its IoU drop to 0.891. This represents the lowest performance among all three training setups for that client. Although FedProx maintains high precision, it lacks the flexibility to capture the full extent of complex salt boundaries in these surveys.The proximal term in FedProx acts as a mathematical anchor that penalizes local updates for deviating from the global model. This constraint restricts the framework from learning survey-specific geological signatures and unique acquisition footprints essential for accurate segmentation. For Client 3, which contains only 1.5% salt features, the μ penalty suppresses the aggressive gradient updates required to identify sparse salt bodies

4.3 Client behavior and variability in federated training

Figure 3 illustrates the training dynamics across 200 communication rounds for the three clients. Client 3 (F3) shows a high initial IoU followed by a sharp dip around round 15. This behavior correlates with its extreme data imbalance of 98.5% background samples. The model initially masters the dominant no-salt class before global aggregation introduces more complex salt features. Client 1 (TGS) and Client 2 (SEAM) provide critical salt-specific information with their higher foreground ratios of 25.0% and 29.0%. These two clients act as the primary knowledge donors for the federation. You can see the global model regularizing the learning process as it stabilizes the performance of all participants. Through collaborative training, Client 1 successfully approaches an IoU of 0.80 despite its smaller sample size. The final convergence proves that the federated framework effectively overcomes extreme class imbalances and domain differences in seismic surveys.

4.4 Qualitative assessment of salt segmentation

Figure 4 presents a qualitative comparison between the original seismic images, ground truth masks, and the predicted masks generated by the TransFed-SaltNet. Each row corresponds to a different test sample, showcasing the model''s performance in delineating salt bodies. For the top two samples, where salt presence is minimal or absent, TransFed-SaltNet correctly predicts empty masks, demonstrating strong precision and the ability to avoid false positives. In the remaining cases, especially where salt structures are

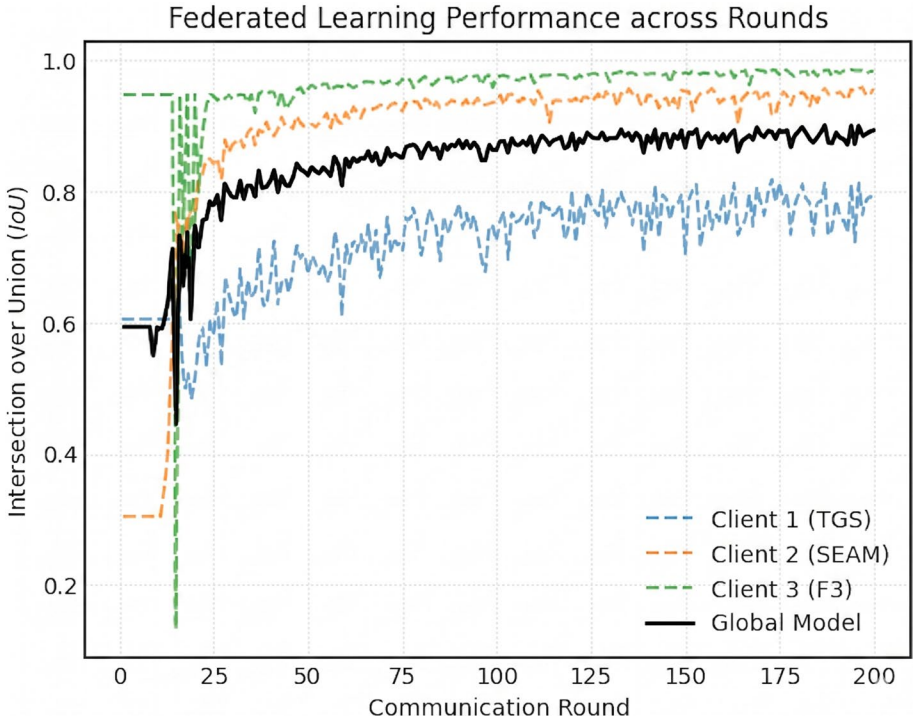


Fig. 3 Global vs. Local Validation IoU scores across 200 federated training rounds for salt body segmentation

more complex, both TransFed-SaltNet exhibit high agreement with the ground truth. The predicted masks from the proposed model closely resemble those from the ground truth counterpart, capturing key salt boundaries with only minor discrepancies at edges or fine structures. These visual results further support the quantitative findings, highlighting the federated model"s ability to generalize well, despite being trained without centralized data aggregation.

4.5 Comparison with prior work

Table 4 compares the performance of GeoFedNet [17] and TransFed-SaltNet on the salt dome segmentation task. Both frameworks achieved high segmentation accuracy, with centralized models performing slightly better than their federated counterparts. GeoFedNet attained an IoU of 0.781 and F1 score of 0.8554 using Fedederated averaging (FedAvg), while TransFed-SaltNet achieved a higher overall performance with an IoU of 0.874 and F1 score of 0.9118 on the testing set. This represents a 9.3% improvement in IoU and a 6% gain in F1 score. The integration of a pretrained EfficientNet-B7 encoder drives these performance gains. The results confirm that transfer learning provides superior performance compared to training models from scratch. This approach significantly enhances both segmentation accuracy and generalization across diverse geological datasets.

4.6 Limitation and future directions

This study identifies several limitations that provide opportunities for future research. The framework currently relies on standard FedAvg, which is susceptible to client drift in extreme non-IID scenarios. Future work should evaluate robust optimizers such as FedNova [24] or SCAFFOLD [25] to mitigate this effect. While the model ensures data

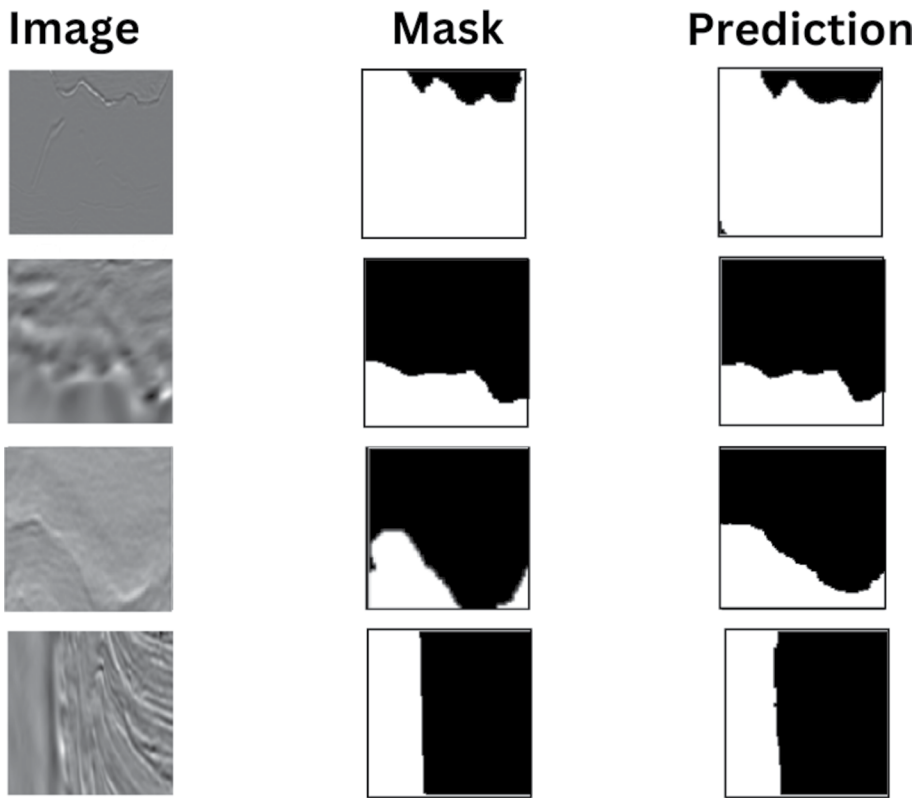


Fig. 4 Qualitative comparison of ground truth and federated model predictions. Each row shows the original seismic image from different datasets used, its ground truth (actual salt mask), and the predicted mask. The results demonstrate that the federated model performs comparably to the actual masks, accurately segmenting salt bodies across various geological structures

Table 4 Performance comparison of GeoFedNet and TransFed-SaltNet models on salt dome segmentation

Model	Setup	Transfer learning	IoU	F1-score	Accuracy
GeoFedNet	Centralized	No	0.848	0.9005	0.9685
GeoFedNet	Federated	No	0.781	0.8554	0.9525
TransFed-SaltNet	Centralized	Yes	0.905	0.9480	0.9820
TransFed-SaltNet	Federated	Yes	0.874	0.9118	0.9726

isolation, it does not yet include formal privacy-preserving layers. Integrating Differential Privacy or Secure Aggregation protocols is necessary to prevent potential model inversion attacks.

A significant challenge remains the extreme foreground-background class imbalance observed across the surveys. For instance, the F3 dataset contains only 1.5% salt pixels, which restricts the model from fully learning sparse geological features. Future research should investigate cost-sensitive loss functions or specialized sampling strategies to address this imbalance within a federated context.

This framework utilizes ImageNet-pretrained weights for encoder initialization. These weights reflect natural image statistics that differ significantly from seismic amplitude patterns. Future research should explore seismic-domain pretraining using self-supervised frameworks like [26]. Implementing adaptive techniques such as [27] will further enhance segmentation accuracy across diverse geological surveys.

End-to-end fine-tuning creates substantial communication overhead. The 66-million parameter EfficientNet-B7 backbone demands heavy data transfers at 32-bit precision. A single payload requires 264 MB. Updating three clients requires both uplink and downlink transfers, costing 1.58 GB per communication round. Over 250 rounds, the total training volume reaches exactly 396 GB. This massive network throughput challenges bandwidth-constrained environments. Future work could explore weight quantization and partial network freezing to reduce this cost while maintaining model performance.

Additionally, this single-machine simulation does not account for real-world deployment issues like network latency, limited bandwidth, or client dropout. Finally, testing the framework on entirely independent geological basins will be essential to verify its out-of-distribution generalization.

5 Conclusion

This study proposed a federated transfer learning framework, TransFed-SaltNet, for salt body segmentation that addresses labeled data scarcity and generalizability challenges by enabling collaborative model training without requiring raw data exchange between organizations. Through extensive experiments, the proposed model achieved strong performance on both validation and held-out test data, demonstrating an IoU of 0.877 and an F1-score of 0.9220 on validation, only marginally lower than the centralized model. When evaluated on an unseen held-out data, the model maintained a high IoU of 0.874 and an F1-score of 0.9118, further confirming its generalization capability across diverse geological settings. These results suggest that federated learning can serve as an effective and privacy-preserving alternative to centralized approaches in seismic interpretation workflows.

This study demonstrates that federated learning is not only technically feasible but also strategically advantageous for real-world seismic segmentation, offering a promising path forward for secure, scalable, and collaborative geoscientific AI. Future work will focus on extending this framework through personalized federated learning, client weighting strategies, and domain adaptation to better handle data imbalance and variability. Expanding its application to other subsurface interpretation tasks beyond salt body segmentation, such as fault detection and horizon tracking, can further enhance its utility in practical geoscience workflows.

Author contributions
M.S.I: Conceptualization, Experimentation, Writing A.W: Writing, Review, Editing, Supervision

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Data availability
The datasets analysed during the current study are available at https://www.kaggle.com/competitions/tgs-salt-identification-challenge and https://tinyurl.com/seamf3.

Code availability
Code is publicly available through GitHub repository https://github.com/saif93/federatedtransferlearning.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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